

DDoS Attack Analysis

Summary	The organization experienced a denial-of-service attack that caused a two-hour network outage. Service was restored by blocking incoming malicious packets and temporarily disabling non-critical network services.
Identify	A hacker targeted our company with an ICMP packet flood attack. All critical network services needed to be restored
Protect	Configured the firewall to block incoming ICMP packets, cutting off the flood traffic at the network perimeter.
Detect	The cybersecurity team configured source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets and Implemented network monitoring software to detect abnormal traffic patterns.
Respond	For future incidents, the plan is to isolate affected systems to contain disruption, prioritize restoring critical services, and review network logs for additional suspicious activity. Incidents will also be reported to upper management and, where applicable, the appropriate legal authorities
Recover	To recover from a DDoS attack by ICMP flooding, access to network services needs to be restored to a normal functioning state. In the future, external ICMP flood attacks can be blocked at the firewall. Then, all non-critical network services should be stopped to reduce internal network traffic. Next, critical network services should be restored first. Finally, once the flood of ICMP packets have timed out, all non-critical network systems and services can be brought back online.

Reflections/Notes: This exercise reinforced how each NIST function builds on the last: identifying the attack vector (ICMP flooding) directly informed the firewall rule used to protect the network, which in turn shaped the detection logic needed to catch similar attempts earlier next time. It also highlighted the value of having a clear, ordered recovery plan rather than restoring everything at once, since prioritizing critical services first minimizes business impact during the recovery window.